



Critical role of single drop microextraction for drug isolation from complex matrix towards efficient pharmaceutical analysis: Advances and challenges

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ABSTRACT

The accurate quantification of drug analytes in complex matrices, such as serum, urine, and pharmaceutical formulations, depends heavily on efficient sample preparation techniques to eliminate matrix interference. Analytical instruments often face limitations in handling raw matrices, making effective pretreatment a prerequisite for precise and reproducible results. While recent trends like miniaturization, automation, high-throughput systems, and reduced solvent usage have advanced sample preparation, they often fall short of addressing the intricacies of complex matrices. Single-drop microextraction (SDME) has emerged as a powerful, eco-friendly alternative for isolating analytes using minimal organic solvent volumes. Its compatibility with advanced analytical techniques such as HPLC, GC, CE, and spectrophotometry makes it a versatile pharmaceutical analysis tool. This review examines the theoretical foundations and practical applications of SDME, highlighting its role in isolating drugs from biological fluids, pharmaceutical formulations, and environmental samples. The classification of topics discussed includes the analyte isolation in various sample types, the impact of matrix effects on drop stability, and the integration of artificial intelligence in optimizing SDME processes. This review aims to comprehensively understand SDME's potential in modern pharmaceutical analysis by presenting a structured overview.

1. Introduction

Precisely determining the amount of a constituent in a sample is essential across diverse sectors such as industry, environment, food, and pharmaceuticals [1,2]. Ensuring the precision and dependability of analytical outcomes relies significantly on effective sample preparation (SP) [3,4]. The elimination of interferences from the sample matrix, boosting analysis sensitivity, and improving overall analytical efficiency [5–7]. Additionally, proper SP contributes to prolonging the lifespan of analytical instruments by averting damage caused by contaminants in the sample [8].

Depending on the sample and target analyte, diverse SP methods are available. Conventional SP methods like liquid-liquid and solid-phase extraction (SPE), designed to isolate organic compounds from aqueous solutions, suffer drawbacks such as being time-intensive and requiring substantial amounts of toxic organic solvents (OSs) [9,10]. Emerging methods like microextraction strive to minimize solvent usage, decrease

sample volume, and simplify extraction. Technological progress has also introduced automated SP systems, enhancing efficiency and reproducibility [11]. Notably, solid-phase microextraction (SPME) and liquid-phase microextraction (LPME) stand out among these advancements [12]. In essence, contemporary SP techniques aim to minimize environmental impact, lower costs, and enhance efficiency without compromising the accuracy and reliability of analytical results [13–15].

LPME, as illustrated in Fig. 1(A), encompasses four primary modes. Also, various types of LPME have been developed to enhance extraction efficiency, selectivity, and environmental sustainability, each offering unique advantages for sample analysis [16]. Cold-induced aggregation microextraction (CIAME) leverages temperature variations to concentrate target analytes onto either a solid or liquid phase. Electro membrane isolation (EMI) utilizes an electric field to preferentially gather charged species. With distinctive solvent characteristics, ionic liquids (ILs) are applicable in microextraction for a diverse array of analytes. In situ, solvent formation microextraction (ISFME) involves the creation of

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a solvent microdroplet within the sample to extract dissolved substances. Liquid-gas-liquid microextraction (LGLME) employs a floating liquid microdroplet to facilitate the extraction of analytes situated between two immiscible liquid phases. Additionally, microwave (MW)-assisted microextraction improves extraction efficiency by utilizing microwave irradiation to expedite mass transfer. SBME utilizes a coated stir bar to selectively extract analytes from the sample matrix. Each of these microextraction methodologies serves as a robust instrument for trace analysis, characterized by high sensitivity, selectivity, and reduced sample preparation demands [17,18].

Furthermore, LPME can function in both static and dynamic modes, with the potential for automation to enhance efficiency, precision, and throughput. In static LPME, the extraction solvent is maintained in a syringe or on a microfiber without movement. Conversely, in dynamic LPME, the extraction solvent is continuously circulated through the sample matrix or vice versa, improving the mass transfer rate and shortening extraction duration. The automation of LPME presents considerable benefits regarding efficiency, precision, and throughput, establishing it as an essential tool in contemporary analytical laboratories [22].

SDME is a technique in which a single drop of OS is used to extract the analytes. SDME encompasses six modes, including direct immersion (DI), SDME; HS-headspace, SDME; 3P—three-phase, SDME; BD-bubble-in-drop, SDME; DD-drop-to-drop microextraction; CF—continuous-flow microextraction [20]. A schematic diagram of LPME and Mode of SDME is demonstrated in Fig. 1(B). SDME extracts are analyte-rich and purified from endogenous compounds, reducing solvent usage by 99 % compared to traditional liquid-liquid extraction. Moreover, there is no need to evaporate the solvent and reconstitute analytes before injection [23]. All SDME techniques are coupled with analytical instruments, separation techniques such as chromatography or CE, and detection techniques such as spectroscopy.

Combining SDME techniques with additional methods like

ultrasound can significantly enhance their performance and sensitivity. These hybrid approaches integrate SDME with methods such as ultrasound, microwave, and other liquid-phase microextraction (LPME) techniques to improve mass transfer, reduce extraction time, and enhance analyte recovery. For example:

Ultrasound-Assisted SDME: SDME is combined with ultrasonic irradiation to enhance analyte diffusion from the sample matrix into the extraction droplet. Ultrasound generates acoustic cavitation, which disrupts the sample matrix, increases mass transfer, and accelerates equilibrium. UA-SDME is particularly useful for complex matrices such as biological and environmental samples, where analytes may be tightly bound. The method improves extraction efficiency while maintaining low solvent consumption.

Also, in MW-SDME, microwave radiation is applied to the sample to provide rapid heating, which enhances analyte solubility and diffusion into the extraction droplet. This technique is particularly useful for solid or semi-solid samples, where microwave energy can break down sample matrices and release target analytes more effectively. MW-SDME offers rapid extraction with reduced solvent usage and improved sensitivity.

By combining SDME with these advanced techniques, the extraction process can be tailored to specific analytes and sample matrices, improving sensitivity, selectivity, and environmental sustainability while maintaining the simplicity and cost-effectiveness of SDME.

Direct immersion single-drop microextraction (DI-SDME) involves immersing a microdrop of water-immiscible organic solvent (OS) into an aqueous sample solution (SS) for analyte extraction, which is then analyzed using techniques like CE, HPLC, or GC [24–27].

Drop-to-drop SDME, a scaled-down version of DI-SDME, uses microliter volumes of OS and sample, achieving quicker equilibrium without stirring but with a lower enrichment factor [28]. In **bubble-in-drop SDME**, introducing air into the syringe increases the droplet's surface area and extraction efficiency [29].

Continuous flow SDME exposes the OS droplet to a circulating SS,

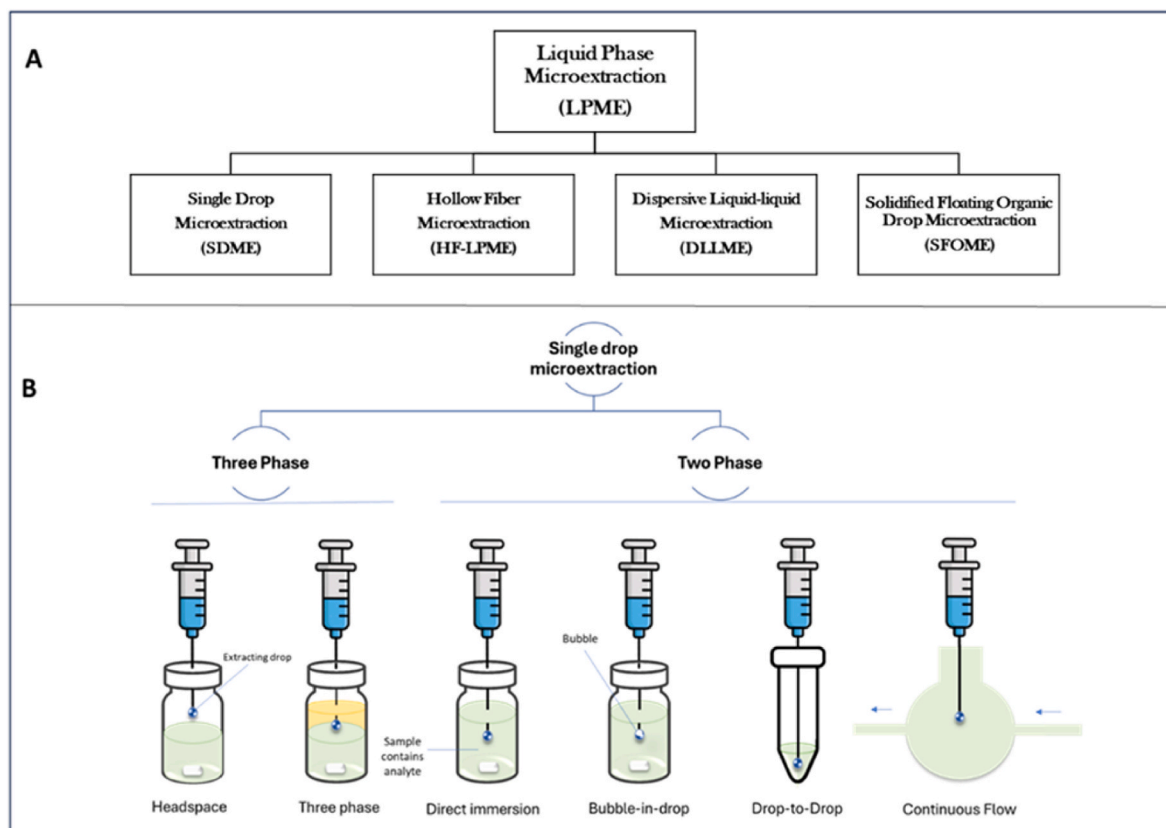


Fig. 1. (A) Schematic diagram of LPME (adapted from Ref. [19]), and (B) Mode of SDME (adapted from Refs. [20,21] and Re-Draw by CorelDraw 5.6 software).

enhancing extraction efficiency through convection and diffusion, though it is limited to non-polar and semi-polar compounds due to solvent compatibility [30].

Liquid-liquid-liquid microextraction involves transferring the analyte through three phases: from the SS to an OS layer in its non-ionic form and then into an aqueous phase where it becomes ionic again, with pH adjustments tailored to the analyte's properties [31].

Headspace (HS-SDME) is a highly utilized technique for matrix elimination and is even more prevalent than DI-SDME [32,33]. In this method, a droplet of the solvent is exposed to the HS of the heated SS to interact with the volatilized analyte [5]. HS-SDME offers several advantages, including enhanced selectivity and the capability to eliminate interferences from complex or dirty matrices and particulate matter [34, 35].

Stirring the SS speeds up the equilibrium of the target analytes from the aqueous phase to the organic drop [36,37]. However, a high stirring rate can cause the micro-drop to detach from the tip of the micro-syringe needle during the extraction process in conventional SDME [38]. HS-SDME provides greater flexibility in terms of the sample stirring rate and the choice of OSs, as it is not constrained by these factors [39]. In this method, the selection between DI-SDME and HS-SDME methods is based on the properties of the analyte and the matrix [40]. Indeed, DI-SDME is better suited for high molecular weight, non-polar to mid-polar analytes. On the other hand, HS-SDME is suitable for semi-volatile and volatile polar, as well as non-polar, low-molecular-weight compounds [34,41]. However, SDME has drawbacks, including drop dislodgement, solvent evaporation, and partial miscibility with water because the solvent drop is suspended at the needle tip. These issues are further exacerbated in GC-based analysis, which requires low-boiling and low-viscosity SDME solvents. Moreover, there is limited scope for adjusting stirring rates and temperature variations [42].

Analytical chemists highly value Single drop microextraction (SDME) for its simplicity, cost-effectiveness, and environmental sustainability, as it requires minimal organic solvents and readily available equipment. It offers high sensitivity through significant analyte pre-concentration and is versatile across various compounds and matrices. By combining extraction, concentration, and sample introduction in a single step, SDME enhances analytical efficiency. However, it has limitations, such as microdrop instability at high stirring speeds, restricted drop volumes (~5 μ L), and challenges in achieving consistent drop size, which can impact reproducibility and efficiency. Optimizing parameters like solvent type, drop volume, stirring rate, and extraction time is essential for reliable results [43,44]. The summary of the advantages and disadvantages of the SDME, as illustrated in Fig. 2, is presented.

Various factors, including temperature, pH, ionic strength, agitation, suitable solvent, extraction duration, and drop size, impact the efficiency of SDME [45]. Choosing an extraction solvent is one of the critical factors in SDME. To improve selectivity and sample matrix elimination in SDME, using a mixture of various OSs or incorporating additives like ion-pair, chelating, and chiral agents into the extraction drop can be beneficial. This approach has the potential to enhance selectivity and expand the range of applications for SDME [46,47].

In SDME, the ion-pair technique is appropriate for polar analyte separation, in which the polar analyte reacts with charged species in an OS and produces a natural compound that has more solubility in a drop. An analyte can be extracted through this procedure and extraction efficiency increases. In other words, in a two-phase system, an analyte polar agent links to lipophilic counterions transforms to a natural form, passes a liquid-liquid border, transfers to a drop (phase-transfer step), and is extracted. This technique increases precision and selectivity [48–51]. This technique allows chelating agents to create stable complexes from analytes that enhance inorganic compound extraction, especially metal ions. This method also increases selectivity [52,53]. Chiral agents have been used for the separation of chiral and enantiomeric molecules from diastereomers that have different solubility in OSs, causing selective extraction [54]. It is important to consider that the chosen solvent for HS-SDME should not be excessively volatile, as evaporation occurs more rapidly in the HS than when the drop is submerged in the sample. However, water-based solvents can still be used in HS-SDME [55]. The solvent used in HS-SDME must meet specific criteria to ensure optimal performance. It should possess a high boiling point and low volatility to minimize evaporation during extraction and maintain stability while also exhibiting chemical stability to prevent unwanted reactions or degradation.

Additionally, the solvent must demonstrate high extraction efficiency, enabling maximum recovery and sensitivity of target analytes. It should form a stable droplet at the needle tip, indicating strong cohesive and adhesive properties crucial for successful extraction. Furthermore, the solvent should not interfere with separating or detecting naive peaks during subsequent analysis, allowing for accurate quantification [56–58]. Traditional solvents such as 1-octanol, 1-butanol, ethylene glycol, toluene, and longer-chain hydrocarbons like n-decane are commonly used in HS-SDME due to their favorable properties and extraction efficiency [33].

The extraction solvent in DI-SDME must have some special characteristics, including 1. low water solubility to extract analytes well, 2. good drop stability when stirring, and 3. low level of toxicity [59,60]. Toluene is a common solvent in DI-SDME because of the good solubility

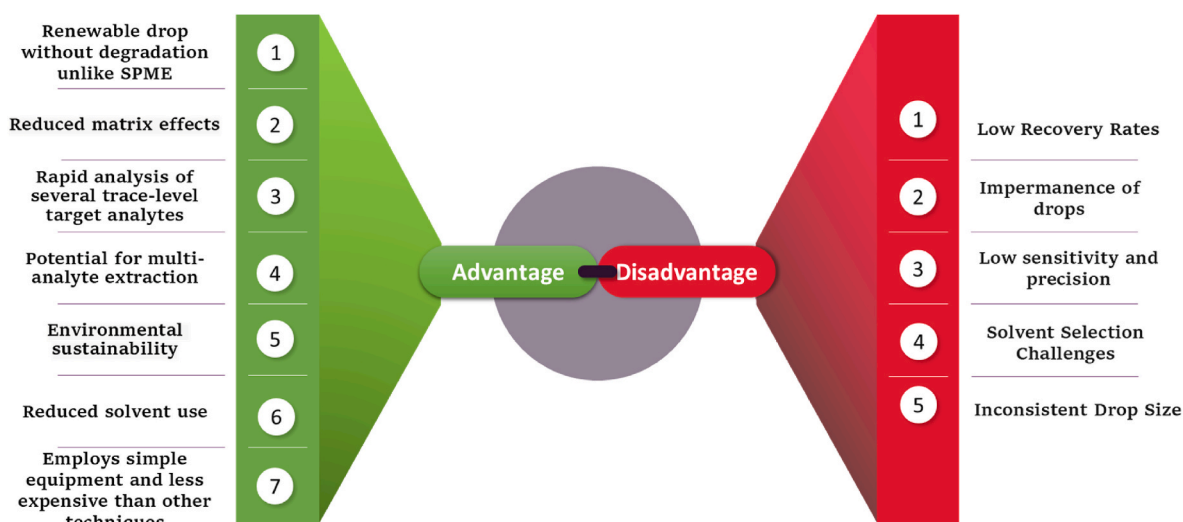


Fig. 2. illustrates the summary of SDME pros and cons.

of analytes in Toluene, and it is immiscible in aqueous solutions [61]. Furthermore, the exploration of novel solvents, such as ILs or their combinations with conventional solvents, continues to broaden the possibilities in SDME. These advancements in solvent selection can enhance extraction efficiency and selectivity in SDME techniques [62, 63]. After extraction, using analytical instruments such as HPLC and GC is crucial, and for this purpose, sometimes a derivatization process is needed.

Derivatization in chromatographic analysis is primarily used to improve analyte separation, thermal stability, and detection sensitivity. It converts analytes into volatile compounds for GC analysis with detectors like FID, ECD, or MS or into fluorescent compounds for HPLC with fluorescence detection. In GC, polar analytes are derivatized to reduce polarity and enhance detector sensitivity, while in HPLC, derivatization is essential for analytes lacking chromophores to enable sensitive detection. This technique enhances sensitivity, selectivity, extraction efficiency, and data quality, with the added advantage of allowing extraction, concentration, and derivatization in a single step [64–67].

In pharmaceutical analysis, chemical derivatization is often needed to enhance the sensitivity and/or the selectivity of the overall analytical process, particularly when using LC. Derivatization generally involves extra sample handling aimed at removing unreacted reagents or pre-concentrate the derivatives [68]. There are relatively few publications on SDME extraction and derivatization. However, there is some notable research, including the determination of free and homocysteine thiolactone (HTL) in human urine [69] and blood analysis [70–72].

These studies showcase the potential of incorporating derivatization or complexation reagents into the extraction process to enhance selectivity and reduce sample background effects for specific analytes. Furthermore, in addition to in-drop derivatization [73], in-syringe derivatization after SDME extraction has also been reported for the analysis of sulfonamides in urine samples [74]. This approach allows for the derivatization step to be performed within the syringe, providing further opportunities for SP and analysis. Overall, the incorporation of derivatization or complexation reagents in SDME techniques offers improvements in the selectivity and sensitivity of the analysis, enabling the determination of a wider range of target analytes in various sample matrices [75,76].

This review fills a gap in the literature by providing a comprehensive

discussion on analyte extraction in pharmaceutical analysis using SDME. It covers the theoretical principles of SDME, its applications in human biological fluids, pharmaceutical samples, and environmental water, and strategies to address challenges such as drop stability and extraction efficiency. The review also explores the use of artificial intelligence to optimize SDME processes and interference elimination. A historical diagram (Fig. 3) illustrates the evolution of SDME in pharmaceutical analysis, and the paper concludes with an overview of current trends, challenges, and future directions, offering a valuable resource for researchers and practitioners.

2. Pharmaceutical analysis in human biological fluids

Biological fluids represent complex samples containing a mixture of proteins, phospholipids, inorganic salts, and organic compounds. These fluids include plasma, serum, cerebrospinal fluid, and urine, among others. Consequently, direct handling of these matrices by chromatographic equipment is not feasible for pharmaceutical analysis [77,78].

Urine stands out as one of the most extensively studied biological matrices for drug analysis (DA) due to its relatively easy collection process and its role as a primary route for the excretion of parent drug compounds and metabolites. It is characterized by moderate complexity, typically containing both organic and inorganic components, along with a relatively high salt content. The efficiency of drug extraction from urine can vary due to various factors. Urinary pH changes can influence drug elimination and extraction efficiency, particularly if they significantly impact the drug's acid/base balance. Variations in ionic strength can also have a significant effect on drug extraction [79–82]. Other biological fluids, such as saliva, plasma, and breast milk, require more complex techniques due to their higher complexity. For example, Wu et al. [83] compared SDME with micro-volume pipette extraction coupled with capillary electrophoresis (CE) for analyzing five tricyclic antidepressant drugs (TCAs)-nortriptyline, imipramine, trimeprazine, promethazine, and iminodibenzyl-in spiked urine samples. Their findings identified toluene as the optimal solvent for efficient SDME of TCAs, highlighting its compatibility with the hydrophobic nature of these analytes. While the study demonstrated the potential of SDME for selective extraction, its application to real-world urine samples with complex matrices would require further validation to assess reproducibility and matrix effects. Additionally, the use of toluene raises concerns

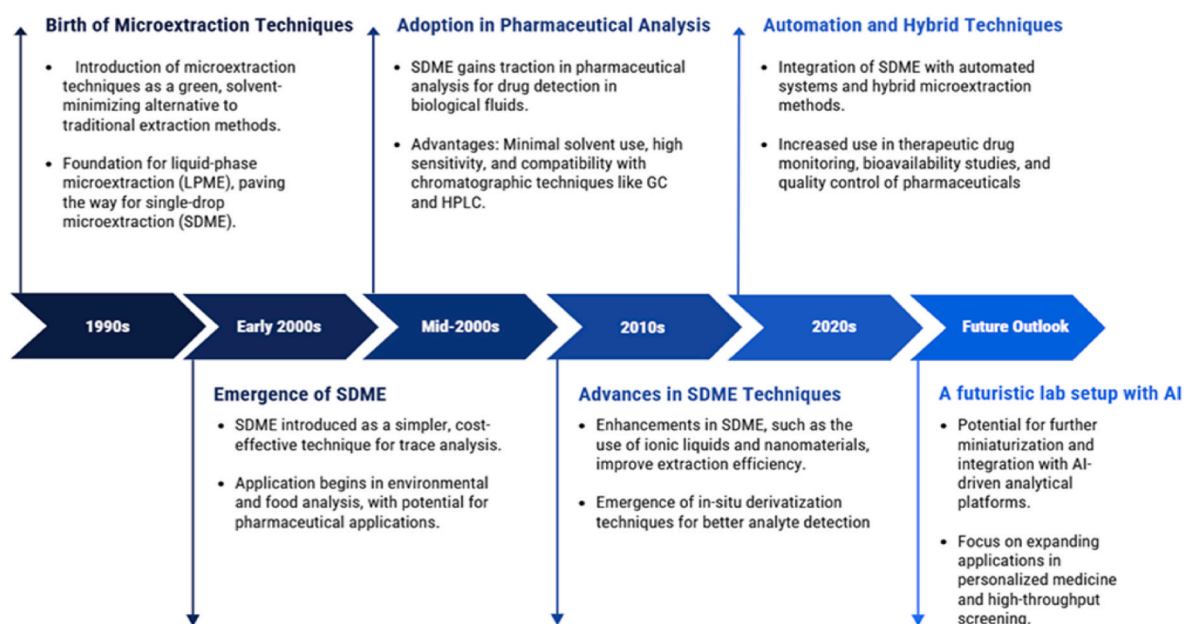


Fig. 3. The development history of pharmaceutical analysis using SDME.

regarding solvent toxicity and environmental impact, suggesting the need for greener alternatives in future studies. Also, Shrivastava and co-workers [84] introduced an innovative approach by coupling atmospheric pressure matrix-assisted laser desorption/ionization mass spectrometry (AP-MALDI-MS) with SDME for the rapid screening of low molecular weight drugs (<500 Da) in human urine. This study showcased the versatility of SDME in facilitating direct coupling with advanced detection techniques. However, the method primarily focused on low molecular weight drugs, leaving room for exploration of its applicability to larger or more structurally complex compounds. The integration of AP-MALDI-MS with SDME represents a step forward in high-throughput analysis, but further optimization of extraction conditions could enhance sensitivity and minimize potential ion suppression effects during MS analysis. Interestingly, Choi and colleagues [85] applied three-phase SDME coupled in-line with chiral CE for the enantioselective analysis of weakly basic amine compounds such as amphetamine directly from urine samples. This approach demonstrated the dual benefits of matrix isolation and preconcentration with minimal sample pretreatment. The incorporation of chiral analysis is particularly noteworthy, as it addresses the growing demand for enantiomer-specific pharmaceutical monitoring. However, the study's reliance on CE as the separation technique might limit its applicability to broader analytical scenarios where higher sample throughput or robustness is required. Future work could explore the integration of three-phase SDME with

alternative separation platforms, such as liquid chromatography, to enhance versatility. Also, Poliwoda et al. [86] developed a fully automated SDME-CE system for the in-line preconcentration and determination of muscimol (MUS) and psilocin (PSC) in urine samples. The method achieved detection limits of 0.004 mg/L and 0.016 mg/L for PSC and MUS, respectively, demonstrating excellent sensitivity. Automation of SDME is a significant advancement that addresses issues of reproducibility and operator variability in manual extractions. However, the study's focus on a narrow range of analytes limits its generalizability. Expanding the system to accommodate a broader spectrum of pharmaceuticals and validating it with real clinical samples would further strengthen its utility for routine analysis. Importantly, García-Vázquez et al. [87] employed three-phase SDME coupled with CE for the determination of nonsteroidal anti-inflammatory drugs (NSAIDs) in urine. The use of ethyl acetate as the organic phase (OP) resulted in enrichment factors up to 44-fold and detection limits ranging from 1.0 to 2.5 µg/mL. This study highlighted the effectiveness of SDME in enhancing the sensitivity of NSAID detection. However, the relatively limited enrichment factors for certain compounds (e.g., 12-fold for ibuprofen) suggest a need for solvent system optimization to achieve uniformly higher preconcentration efficiencies. Furthermore, while valuable for small-scale applications, the reliance on CE may face scalability challenges in high-throughput clinical settings. In another work, Chinese scientists Ma et al. [88] investigated the use of *o*-dibutyl phthalate as an

Table 1

Technical and analytical figure of merit for the pharmaceutical analysis based on matrix elimination by SDE method.

Analytes	OP	Matrix	Method	Sep/Det	LOD	Linear range	Ref.
Nortriptyline, Imipramine, Trimeprazine, iminodibenzyl and Promethazine	Toluene	Urine	DI-SDME	CE	0.15–50.0 mg/mL	2.50–62.50 mg/mL	[83]
Nortriptyline, Amitriptyline, Imipramine, Trimeprazine and Quinine	Xylene	Urine	DI-SDME	AP-MALDI-MS	0.3–1.6 µM	–	[84]
Amphetamine, <i>p</i> -Chloroamphetamine	Octanol	Urine	SDME	CE	2–5 nM	10–3000 nM	[85]
Muscimol and Psilocin	Octanol	Urine	SDME	CE	0.004–0.016 mg/L	0.05–50 mg/L	[86]
Ibuprofen, Naproxen, Diclofenac Sodium, and Ketoprofen	Ethyl acetate	Urine	SDME	CE	1.0–2.5 µg/mL	2.5–100 µg/mL	[87]
Lidocaine, Bupivacaine, and Tetracaine	<i>o</i> -Dibutyl phthalate	Urine	DI-SDME	HPLC-UV	Lower than 0.05 µg/ml	0.1–10.0 µg/mL	[88]
Sufentanil and Alfentanil	Octanol	Urine	DI-SDME	GC-FID	5–6 ng/mL	50–1000 ng/mL	[89]
Nicotine, Anabasine, and cotinine	Chloroform	Urine, saliva	DI-SDME	GC-FID	0.33–0.45 mg/L	0.5–25.0 mg/L	[90]
Aminophenol isomers (APs)	1-hexyl-3-methylimidazolium	Urine, hair	SDME	HPLC-UV	0.6 to 0.14 µg/mL	0.21–91.4 µg/mL	[91]
Chloroquine	Octanol	Urine	LLLME	Isocratic RP-HPLC-UV	0.3 µg/L	1–200 µg/L	[92]
Amphetamine (AP), Methamphetamine (MAP)	<i>n</i> -hexane	Urine	LLLME	HPLC-UV	0.5 µg/L	1–1500 µg/L	[93]
Captopril	Gold nanoparticles	Human serum, pharmaceuticals	HS-SDME	microchip-photothermal lens microscopy	0.31 nM	0.5–70 nM	[94]
Hyperforin, Hypericin, Pseudohypericin	<i>n</i> -octanol and chloroform mixture	Plasma, urine	SDME	HPLC-UV-FLD	1.5–6.0 ng/ml	130–290 ng/mL	[95]
Valproic acid	<i>n</i> -dodecane	Human blood serum	HS-SDME	GC-FID	0.8 µg/ml	2–20 µg/mL	[96]
Mirtazapine, Venlafaxine, Escitalopram, Fluoxetine, and Fluvoxamine	3-pentanol	Human milk, urine	LLLME	HPLC-UV	0.11–0.24 ng/mL	0.8–400.0 ng/mL	[97]
ketoprofen, Ibuprofen, and Naproxen	Octanol	urine	three-phase SDME	CE-UV	2–10 µg/L	10–7000 nM	[98]
Amphetamine, Methamphetamine, 3,4-Methylenedioxy-Amphetamine-Methamphetamine	Chloroform	Urine	DI-SDME	GC-(PDHID)	0.015–0.5 µg/mL	–	[99]
Amitriptyline and Nortriptyline	Toluene	Urine	DI-SDME	GC-FID	0.04–0.05 µg/mL	–	[100]
Dopamine	Octanol	Urine	DI-SDME	AP-MALDI-MS	40 mg/L	–	[101]
Gatifloxacin (GAT), lomefloxacin (LOM), enoxacin (ENO), ciprofloxacin (CIP), ofloxacin (OFL)	DCM: Toluene	Urine	SD-LLLME	CE	7.4–31.5 µg/L	40–1000 µg/L	[102]

extraction solvent for isolating local anesthetics—lidocaine, bupivacaine, and tetracaine—from human urine. The method achieved notable enrichment factors exceeding 86-fold under optimized conditions, with detection limits below 0.05 µg/mL. This study underscores the potential of SDME for high enrichment and clean-up efficiency within a short extraction time (30 min). However, the selection of *o*-dibutyl phthalate raises concerns due to its potential as an endocrine disruptor, highlighting the need for safer, non-toxic alternatives. Additionally, the lack of detailed evaluation of matrix interferences limits the study's applicability to complex biological samples. To complete our discussion in this sub-section, Table 1 [83–102] summarizes recent reports on matrix elimination in DA in human biofluids in terms of the type of drug, extraction strategy, applied OPs, and biofluid samples.

One of the critical insights from these studies is the optimization of extraction conditions, such as the choice of solvent, which significantly influences the efficiency of drug extraction. For instance, Wu et al. identified toluene as optimal for tricyclic antidepressants, highlighting the solvent's role in enhancing selectivity and sensitivity. This solvent specificity not only improves the extraction efficiency but also reduces matrix interferences, a common challenge in biological sample analysis.

In the studies by Wu et al. [75] and Ma et al. [80], optimal solvents such as toluene and *o*-dibutyl phthalate are identified; however, concerns regarding their toxicity and environmental impact are raised. It is crucial to find greener alternatives that are equally effective. Artificial intelligence, especially machine learning (ML), can be utilized by training on datasets that include solvent properties—such as polarity, viscosity, density, and toxicity—as well as analyte characteristics like hydrophobicity and molecular weight. This approach can help predict the best solvent for SDME for a specific analyte or class of analytes, moving beyond simple trial-and-error methods. Additionally, AI models can be trained to anticipate matrix effects based on the composition of the sample matrix (e.g., components in urine) and the properties of the analyte.

Moreover, integrating SDME with advanced analytical techniques like capillary electrophoresis (CE) and mass spectrometry (MS) showcases a synergistic approach where SDME acts not only as an extraction method but also as a sample clean-up step. This integration is particularly evident in the work by Choi et al., where SDME was in-line coupled with chiral CE for direct analysis of amines from urine, minimizing sample preparation steps and reducing analysis time, which is crucial for high-throughput screening.

The development of three-phase SDME systems, as seen in the research by García-Vázquez et al., further extends the capability of SDME by allowing for the extraction of both neutral and ionized forms of drugs, thus broadening the scope of analytes that can be effectively isolated and quantified. This advancement is significant as it tackles the challenge of extracting compounds with varying physicochemical properties from complex matrices.

Lastly, the automation of SDME, as demonstrated by Poliwođa et al., marks a significant leap toward routine application in clinical settings. Automation enhances reproducibility and reduces human error, making the technique more reliable for quantitative analysis.

In summary, the evolution of SDME from a simple extraction method to a sophisticated, integrated analytical tool in pharmaceutical analysis reflects its critical role in overcoming the challenges posed by complex matrices. Its ability to provide high enrichment factors, low detection limits, and significant sample clean-up within short extraction times positions SDME as an indispensable technique for future advancements in drug analysis.

Also, the analytical performance of the methods was compared based on LOD and linear range. The analytical results highlight the effectiveness of various techniques for overcoming matrix interference in the pharmaceutical analysis of human biological fluids. Among these, SDME stands out as a versatile and efficient method, offering high sensitivity, significant matrix clean-up, and low solvent consumption. Studies reveal its adaptability in coupling with separation and detection techniques. A

general summary shows that CE, GC-FID, and HPLC-UV were the most common coupling techniques for biological analysis with SDME. GCis is effective for volatile and semi-volatile compounds, such as nicotine metabolites [90] and amphetamines [99].

HPLC is widely applied for non-volatile drugs, often in combination with UV or fluorescence detection, as seen in the analysis of antidepressants, fluoroquinolones, and hypericin derivatives. Also, CE shows advantages such as low sample consumption and high separation capability, according to research; however, its limitations in sensitivity and efficiency for non-polar compounds underscore the need for robust preconcentration methods. The obtained limits of detection (LOD) are notably low in the nanomolar range. The analytical performance across the methods varies depending on the analyte class and matrix complexity. While NSAIDs such as ibuprofen and ketoprofen showed reliable detection limits (1.0–2.5 µg/mL) with ethyl acetate as the extraction solvent, antidepressants like amitriptyline and nortriptyline reached even lower detection thresholds (0.04–0.05 µg/mL) using toluene in DI-SDME coupled with GC-FID. However, despite these strengths, there are limitations. Methods like SDME struggle with highly polar compounds due to poor partitioning into OPs, while techniques like AP-MALDI-MS, though sensitive, require costly and specialized instrumentation.

The strengths of these techniques lie in their minimal solvent usage, ease of implementation, and compatibility with a wide array of analytes and detection systems. However, their limitations, such as a dependency on optimal solvent selection and challenges with highly complex matrices, need addressing. For example, SDME's simplicity might not suffice for substances requiring extensive preconcentration or cleanup, particularly in biological matrices with interfering compounds. Tailoring solvent properties or employing hybrid techniques could help overcome these challenges and broaden their applicability.

Alternative elimination and detection methods could provide expanded capabilities for drugs not listed or yet to be studied. Techniques like IL-based extractions could enhance the recovery of polar or ionic drugs, while SPME might be effective for protein-bound or hydrophobic compounds. Similarly, integrating headspace SDME with advanced mass spectrometry platforms could address the challenges of detecting volatile or emerging psychoactive substances. Drop contamination can be mentioned among the limitations in biological analysis; however, selecting the headspace mode will be effective for highly complex samples. Three-phase liquid microextraction provides better matrix clean-up, and fewer interfering compounds enter the acceptor phase. These advancements would complement the existing methods, enabling robust, comprehensive analysis of a wider spectrum of analytes.

The solvent is selected based on its compatibility with the analyte and the detection method. High concentration factor values in various studies indicate that solvent selection plays a significant role in concentration and high sensitivity, and it is more critical in complex samples.

3. Analyte isolation in DA in pharmaceutical samples

Accurately detecting the active ingredients, impurities, and residual solvents in drugs is crucial due to their direct impact on human health and the efficacy of the drugs.

3.1. SDME for active ingredients

Deng et al. [103] utilized microwave-assisted extraction (MAE) combined with headspace single-drop microextraction (HS-SDME) and gas chromatography-mass spectrometry (GC-MS) for the isolation and quantification of paeonol, a bioactive compound in traditional Chinese medicines. The selection of 1-octanol as the extraction solvent was based on its high partition coefficient for paeonol, ensuring efficient extraction from complex matrices. While this approach demonstrated simplicity,

environmental sustainability, and compatibility with natural product analysis, the reliance on 1-octanol as a solvent limits its applicability to compounds with similar physicochemical properties, potentially restricting broader use in pharmaceutical analysis. Moreover, the integration of MAE with SDME, though innovative, may pose challenges in terms of reproducibility and scalability when applied to diverse samples or industrial workflows. Future work could focus on optimizing solvent selection and exploring compatibility with other bioactive compounds to address these limitations.

3.2. Determination of iodine in pharmaceutical samples

Das et al. [104] compared SDME with solid-phase microextraction (SPME) for determining iodine in pharmaceutical samples, demonstrating that SDME offers faster mass transfer, enhanced efficiency, and simpler experimental setups. The use of oxidation and iodination reactions to convert iodide into 4-iodo-N, N-dimethylaniline, followed by GC-MS analysis, underscores the versatility of SDME in coupling with advanced analytical techniques. However, while SDME outperformed SPME in this specific application, the need for chemical derivatization introduces an additional procedural step, which may limit the method's practicality for routine analysis. Similarly, Pereira et al. [97] combined HS-SDME with UV-Vis spectrophotometry for iodine determination, achieving an impressive limit of detection (LOD) of 0.69 µg/L using a 2.5-µL drop of N, N-dimethylformamide as the extraction solvent. Although this method is cost-effective and environmentally friendly, the reliance on a microvolume UV-Vis setup may restrict its adoption in laboratories without access to such specialized instrumentation. Overall, these studies highlight the potential of SDME for iodine analysis but reveal the trade-offs between sensitivity, simplicity, and instrumentation requirements, which warrant further exploration.

3.3. Residual solvent analysis in pharmaceuticals

Residual solvents are critical impurities that can affect drug safety, stability, and efficacy. Saraji et al. [105] employed HS-SDME and dynamic headspace liquid-phase microextraction (DHS-LPME) coupled with gas chromatography-flame ionization detection (GC-FID) to quantify thirteen volatile solvents, including methanol, ethanol, acetone, and benzene. These methods offered high sensitivity and selectivity, particularly for volatile organic compounds (VOCs), but the complexity of dynamic LPME setups may limit their routine applicability. Heydari [106] developed a simpler HS-LPME method using n-hexanol as the extraction solvent, showing that solvent selection plays a pivotal role in enhancing sensitivity and selectivity, while Yu et al. [107] introduced multiple headspace SDME (MHS-SDME) with dimethyl sulfoxide (DMSO) for residual solvent extraction, demonstrating improved reproducibility and extraction efficiency. Notably, Michulec et al. [108] achieved exceptional detection limits (0.001 mg/kg for halogen derivatives and 0.3 mg/kg for hydrocarbons) by using benzyl alcohol as the extraction solvent in pharmaceutical oils. These findings underscore the importance of tailoring extraction solvents and techniques to the target analytes and sample matrices. However, challenges remain in terms of balancing extraction efficiency, solvent selectivity, and method simplicity. Wang et al. [109] further contributed by using HS-LPME GC-FID to determine volatile residual solvents (e.g., tetrahydrofuran, methanol, dichloromethane, and ethanol) in pharmaceutical products, achieving excellent linearity (1–1000 mg/L). While these advancements push the boundaries of residual solvent analysis, the scalability and practicality of these methods for routine quality control in pharmaceutical manufacturing require further investigation.

3.4. Impurities in drug substances

Bagheri et al. [110] utilized immersed SDME for the trace

enrichment of formaldehyde in vaccines, achieving high sensitivity through derivatization with the Hantzsch reaction and subsequent extraction into a chloroform micro-drop. This method demonstrated significant analytical potential for monitoring impurities in complex biological matrices, such as vaccines. However, the use of chloroform, a toxic solvent, raises concerns about environmental and safety implications, suggesting that greener alternatives should be explored. Similarly, Li et al. [111] employed derivatization and HS-SDME followed by HPLC-UV for the detection of genotoxic methanesulfonate impurities (e.g., MMS, EMS, and IPMS) in active pharmaceutical ingredients (APIs). The use of room-temperature ionic liquids (RTILs) such as [BMIM]TF₂N significantly enhanced extraction efficiency and exemplifies the growing interest in sustainable and efficient solvent systems. While these approaches demonstrated high sensitivity and selectivity, the need for derivatization and specialized reagents may limit their adoption in high-throughput industrial settings. Future research could focus on integrating greener solvents and automating the extraction process to improve scalability and reduce environmental impact.

In recent years, many articles have been published in the field of analyte isolation in DA in pharmaceutical samples (formulation), as summarized in Table 2.

The strengths and limitations of SDME-based methods and their potential applications in pharmaceutical analysis highlight their significant contributions to ensuring drug safety and quality. Among the notable strengths, these methods demonstrate exceptional sensitivity and selectivity, with detection limits often in the nanogram or microgram range. For instance, the LOD of 0.69 µg/L for iodine and 0.001 mg/kg for halogen derivatives illustrates their capability for trace-level analysis. Additionally, SDME and its variations, such as HS-SDME and HS-LPME, are environmentally friendly due to the use of minimal solvent volumes, typically ranging from 2 to 5 µL, which significantly reduces solvent waste compared to traditional liquid-liquid extraction methods. The adaptability of these techniques to complex pharmaceutical matrices, including traditional Chinese medicines, solid drug products, pharmaceutical oils, and vaccines, further underscores their versatility. Moreover, the simplicity of the SDME setup and its faster mass transfer compared to other techniques, such as SPME, make it a more accessible and efficient option for many applications.

However, despite these advantages, SDME-based methods have certain limitations that must be addressed. Complex pharmaceutical matrices can introduce interferences that negatively affect extraction efficiency and analyte recovery, necessitating additional sample cleanup or optimization steps. Furthermore, many of these methods are tailored to specific analytes, such as formaldehyde, iodine, or methanesulfonates, requiring significant modifications to extend their applicability to other impurities or active ingredients. The reliance on advanced instrumentation, such as GC-MS, GC-FID, or HPLC-UV, also presents a limitation, as these techniques may not be readily available in resource-limited settings. Additionally, the choice of extraction solvent is critical, as it may restrict the method's applicability to certain analytes. For example, solvents like benzyl alcohol or DMSO may not be compatible with all target compounds, requiring method-specific optimization.

Although the reviewed techniques have largely focused on specific analytes, such as methanesulfonates, iodine, formaldehyde, and residual solvents, their potential for broader applications is significant. These methods can be adapted to detect impurities and active ingredients in other pharmaceutical compounds that have not yet been extensively studied. For instance, with the increased use of antiviral drugs like remdesivir or favipiravir, SDME techniques could be optimized to detect related impurities or degradation products. Similarly, hormones such as estrogens and progestins, often used in contraceptives, may be prone to contamination with residual solvents or impurities, making techniques like HS-SDME or IL-IDME suitable for their analysis. In the field of oncology, chemotherapeutic agents such as methotrexate or 5-fluorouracil could benefit from derivatization-SDME approaches for detecting genotoxic impurities. Additionally, macrolide antibiotics like

Table 2

Technical and analytical figure of merit for the DA in a pharmaceutical formulation based on matrix elimination by the SDME method.

Analytes	Sample	Organic phase	Method	Sep/det	LOD	Ref
Paeonol	Traditional Chinese medicine	1-octanol	MAE-HS-SDME	GC-MS	1 µg/ml	[103]
Iodine	Multivitamins/ Nutraceuticals: Cyltabs, Revital Antiseptics/Disinfectants: Workadine, Betadine, Cipladine, Collosal Iodine	Iso-octane	DI-SDME	GC-MS	0.9998–10 ng l ⁻¹	[104]
Iodine	Thyroid Hormone Replacement: Eutirox Iodine and Folic Acid Supplement: Yodocefol Antioxidant/Nutritional Supplement: Dep. Richelet	N,N-dimethylformamide	HS-SDME	Mvol-UV-vis	0.69 µg l ⁻¹	[112]
Residual solvents (methanol, ethanol, acetone, acetonitrile, hexane, ethyl acetate, THF, cyclohexane, benzene, heptane, pyridine, toluene, and DMF)	Levothyroxine, adult cold, and Acetaminophen	N-Octanol	DHS-LPME and HS-SDME	GC-FID	DHS-LPME: 1.9–44 mg L ⁻¹ HS-SDME: 0.3–21 mg L ⁻¹	[105]
Residual solvents (ethanol, ethyl acetate, toluene)	Pharmaceutical products	N-hexanol	HS-SDME	GC-FID	0.2–0.5 µg/ml	[106]
Residual solvents (methanol, ethanol)	Solid drug products	DMSO	MHS-SDME	GC-FID	0.01 µg	[107]
Residual solvents (hexane, benzene, toluene and trichloromethane, 1,1,1-trichloroethane, tetrachloromethane, trichloroethylene, tetrachloroethene)	Pharmaceutical products	Benzyl alcohol	SDME	GC-FID GC-ECD	0.001 mg kg ⁻¹ and about 0.3 mg kg ⁻¹ for halogen derivatives and hydrocarbons	[108]
Residual solvents (tetrahydrofuran, Dichloromethane, methanol, and ethanol)	Hydroxycarbamide	1-octanol	HS-SDME	GC-FID	0.2–2.0 mg/L	[109]
Formaldehyde	DTP and DT vaccines, diphtheria-tetanus antigen	Chloroform	DI-SDME	GC-MS	0.22 ng/L	[110]
MMS, EMS, and IPMS	APIs	[BMIM]TF ₂ N	HS-SDME	HPLC-UV	15 ng ml ⁻¹	[111]

erythromycin or azithromycin, which may contain residual solvents or degradation products, could be efficiently analyzed using HS-SDME coupled with GC-MS. Emerging biopharmaceutical products, including monoclonal antibodies and mRNA-based drugs, also present new challenges for impurity detection, which could be addressed by further developing these methods to achieve the required sensitivity and selectivity. Future research should aim to optimize these techniques for additional drug classes and impurities, explore the use of green solvents to enhance environmental sustainability and integrate automation to improve reproducibility and scalability. Addressing these challenges will enable the analytical chemistry community to further ensure pharmaceutical products' safety, efficacy, and quality.

4. Analyte isolation in DA in water samples

Pharmaceutical products are recognized as emerging contaminants, raising health concerns due to their presence in environmental waters. The primary source of their release is domestic and hospital wastewater, which contains drugs and metabolites formed after consumption and body metabolism, potentially increasing toxicity and emphasizing the importance of their analysis [113–115]. In the research work conducted by Guo et al. [108], a simple, rapid, and environmentally friendly single-drop microextraction (SDME) technique was developed to determine sulfonamides in environmental water. The use of 1-octyl-3-methylimidazolium hexafluorophosphate ([C8MIM][PF6]) as the extraction solvent demonstrated significant advantages, including low limits of detection (LOD, 0.5–1 ng/mL), good repeatability (relative standard deviation, RSD, 4.2–9.9 %), and a wide linear range (1–1500 ng/mL) with excellent linearity ($R^2 > 0.9989$ for all target compounds). Despite these promising outcomes, the method's reliance on ILs raises questions about the environmental sustainability of the extraction solvent itself, given the potential toxicity and limited biodegradability of

some ILs. Future research should focus on identifying greener ILs or alternative solvents that maintain high extraction efficiency while minimizing environmental impact. In this work, AI algorithms could be employed to predict and optimize the selection of ILs based on their physicochemical properties and target analyte characteristics. Instead of relying on trial-and-error or empirical optimization, AI could rapidly screen a vast chemical space of ILs and predict the most effective candidates for sulfonamide extraction from diverse water matrices.

Mosavian et al. [116] introduced an IL-based immersed droplet microextraction (IL-IDME) technique for determining amitriptyline in wastewater, using 1-hexyl-3-methylimidazolium hexafluorophosphate ([C6MIM][PF6]) as the extraction solvent. The method demonstrated good applicability for complex matrices such as hospital wastewater. However, while IL-IDME is innovative, its immersion-based approach may present challenges in handling small sample volumes or achieving high-throughput analysis. Furthermore, the use of hexafluorophosphate-based ILs, which are known for their potential environmental risks, highlights the need for further assessment of the method's sustainability. Addressing these limitations could pave the way for broader adoption of this technique in pharmaceutical and environmental analysis.

Interestingly, a novel technique for the quantitative determination of low concentrations of Monensin in soil, surface water, and human urine has been developed by Ramaiyan Sekar et al. [117] developed a novel SDME technique for the quantitative determination of trace levels of Monensin in soil, surface water, and human urine. The use of chloroform-toluene (1:1, v/v) as the extraction solvent prior to AP-MALDI-MS analysis enabled effective preconcentration of the target analyte. While the study demonstrated the versatility of SDME for diverse matrices, choosing a chlorinated solvent like chloroform raises concerns about toxicity and safety during the extraction process and waste disposal. A shift toward less hazardous solvents or solvent-free

extraction approaches would enhance the method's applicability in routine analysis and compliance with green chemistry principles. Also, Sercan Yıldırım et al. [118] introduced an innovative lab-in-syringe dispersive ionic liquid SDME (DI-SDME) for analyzing fluoroquinolone antibiotics in environmental waters. This method employed natural deep eutectic solvents (NADES) as extractant drops, representing a significant advance in the field by leveraging environmentally benign solvents. The technique achieved enrichment factors ranging from 35 to 45 ng/L and was coupled online with HPLC and fluorescence detection, offering both efficiency and sensitivity. However, the preparation and stability of NADES in varying environmental conditions remain areas for further investigation. Additionally, the method's scalability and reproducibility for large-scale applications should be evaluated to determine its practicality for routine environmental monitoring. Importantly, a new method for separating sulfonamides from water samples was introduced by Furuza Shakirova et al. [74], who proposed a hybrid approach combining low-density solvent-based solvent-demulsification dispersive liquid-liquid microextraction (LDS-SD-DLLME) with SDME for the separation of sulfonamides from water samples. This fast and convenient method addresses some of the limitations of traditional SDME, such as low phase volume and extraction efficiency, by integrating solvent-demulsification principles. While this hybrid approach shows promise in improving extraction performance, its complexity in terms of solvent selection and method optimization may pose challenges for less experienced users. Simplifying the protocol or developing automated systems could enhance its practical utility.

Table 3 summarizes recent reports on the matrix elimination for DA in water samples to complete our discussion in this sub-section [65, 111–115]. Table 3 also shows the analytical performance of various methods of matrix elimination in the SDME technique.

These methods collectively demonstrate significant strengths, including high sensitivity, wide linear ranges, and environmentally friendly solvent use. For instance, the use of ILs and NADES reduces the environmental impact compared to traditional OSs. Furthermore, the adaptability of these techniques to diverse matrices, such as water, soil, and biological samples, underscores their broad applicability for environmental monitoring. However, certain limitations remain. Matrix interference is a persistent challenge, particularly in complex environmental samples like wastewater and soil, requiring additional sample cleanup or method optimization. Additionally, the specificity of many methods to particular analytes necessitates further modifications to extend their applicability to other pharmaceuticals. The reliance on advanced instrumentation, such as HPLC-MS/MS or AP-MALDI-MS, may also limit their accessibility in routine laboratories.

Despite these challenges, the potential for adapting these techniques to other unreported pharmaceutical contaminants is significant. For example, antiviral drugs such as remdesivir and favipiravir, which have seen increased use during recent pandemics, could be targeted using

SDME or IL-IDME. Hormonal drugs, including estrogens and progestins commonly found in contraceptives, as well as chemotherapeutic agents like methotrexate and 5-fluorouracil, represent other promising targets for these methods. Expanding the use of green solvents, such as NADES, and integrating multiple extraction techniques (e.g., SDME with SPME) could further enhance sensitivity and reduce matrix effects. Automation of microextraction techniques could also improve reproducibility and reduce manual labor, making these methods more accessible for widespread application.

In conclusion, advancements in microextraction techniques, particularly those involving SDME, DLLME, and IL-IDME, have made significant contributions to addressing the challenges of pharmaceutical contamination in environmental waters. These methods exhibit excellent analytical performance, with low detection limits, broad linear ranges, and adaptability to complex matrices.

5. Matrix effect on drop hold

The stability of the extraction solvent drop is critical to the efficiency and reproducibility of the SDME method. The sample matrix composition and nature of the SS can significantly influence the drop's stability. The drop can detach or dissolve if it becomes unstable, leading to poor results or method failure. The stability of the extraction solvent drop is critical to the efficiency and reproducibility of the SDME method. The sample matrix—composition and nature of the SS—can significantly influence the drop's stability. If the drop becomes unstable, it can detach or dissolve, leading to poor results or method failure [122].

Various physical, chemical, and interfacial characteristics of the sample matrix influence the stability of droplets during SDME. The primary factors include:

5.1. Viscosity of the sample matrix

Sample matrices with high viscosity (e.g., concentrated biological fluids) can impose shear forces on the droplet during mixing or agitation, thereby heightening the likelihood of droplet dislodgement or deformation. To address this issue, it may be necessary to employ lower stirring speeds or to utilize static extraction methods for these viscous matrices [34].

5.2. Surface tension and interfacial tension

The relationship between the sample matrix and the OS plays a significant role in determining the surface tension at the droplet interface. A decrease in interfacial tension, which can occur due to the presence of surfactants, emulsifiers, or other surface-active agents within the sample, may compromise droplet stability, potentially resulting in detachment or the dissolution of the solvent into the matrix.

Table 3

Technical and analytical figure of merit for the DA in water based on matrix elimination by SDME method.

Analytes	Organic phase	Matrix	Method	Sep/Det	LOD	Linear range	Ref.
Sulfonamides	1-octyl-3-methylimidazolium hexafluorophosphate	Environmental water	SDME	HPLC-MS/MS.	0.5–1 ng/mL	1–1500 ng/mL	[119]
Amitriptyline	1-Hexyl-3-methylimidazoliumhexphate	Wastewater	SDME	LC/UV	0.004 µg/mL	0.01–10 mg/mL	[116]
Monensin	Chloroform-toluene	soil, surface water, and human urine	DI-SDME	AP-MALDI-MS	–	10–100 ng/mL	[117]
Naproxen, Diclofenac, and Ibuprofen	Toluene	Water	DLLME + SDME	HPLC-UV	0.03–0.2 µg/L	0.1–1000 µg/L	[120]
Fentanyl	n-octane	Plasma, wastewater, and urine	LLLME	HPLC-UV	0.1 ng/mL	0.5–1000 ng/mL	[121]
Fluoroquinolones	Natural deep eutectic solvent	Environmental waters	LIS-automated SDME	HPLC-FLD	6–9 ng/L	0.1 and 5 µg/L	[118]
Sulfanilamide, Sulfathiazole and Sulfamethazine	1-octanol	Lake water, fishery water, and wastewater	LDS-SD-DLLME	HPLC-UV	0.22–1.92 µg/L	1.0–500 to 10–500 µg/L	[74]

Therefore, selecting a solvent with suitable interfacial properties, such as immiscibility with the matrix, is essential for preserving droplet integrity [24].

5.3. Matrix composition (surfactants, proteins, and salts)

Surfactants: The presence of surfactants in the sample matrix can lower interfacial tension, thereby destabilizing the droplet. This may also lead to emulsification or partial dissolution of the droplet into the sample.

Proteins: In protein-rich matrices, such as serum or milk, proteins can adsorb at the droplet interface, affecting its stability. Additionally, proteins may contribute to droplet fouling or aggregation.

Salt Concentration (Ionic Strength): Elevated salt concentrations can induce alterations in the solubility of the solvent (salting-out effect), which may either stabilize or destabilize the droplet, contingent upon the properties of the solvent. However, excessive ionic strength can result in droplet shrinkage or detachment.

5.4. pH of the sample matrix

The pH level of the sample can influence the chemical stability of the solvent droplet or facilitate reactions at the interface. For instance, acidic or basic conditions may lead to the degradation of certain OSs or cause ionizable analytes to modify their partitioning behavior [123–127].

Kubasová and colleagues [39] conducted a study to assess the influence of sample matrix on the stability of microdrops in both direct and headspace modes. The investigation encompassed four distinct sample types: aqueous solutions, acidic matrices (specifically orange juice), alcoholic matrices (plum brandy), and protein-rich matrices (milk). The analysis of orange juice revealed two significant challenges: the transfer of pigments from the juice to the micro-drop, which led to instability, and the high fiber content that complicated the extraction process. While increasing the dilution factor enhanced the visibility of the microdrop, inadequate dilution impeded both the observation and retraction of the drop post-extraction. To mitigate this issue, the authors suggested employing a "sample white-out" technique for undiluted samples; however, even the application of a lightning agent did not consistently ensure adequate visibility of the droplet. A notable distinction in the volume of extracted drops was observed between the headspace and direct modes, with the direct mode exhibiting a more significant variation, especially at elevated sample concentrations. In the case of alcoholic samples, the high ethanol concentration restricted both the volume of the droplets and the stirring rate. This limitation was linked to the solubility of the selected extractant solvent (such as toluene) in ethanol, which diminished the overall droplet volume. In matrices rich in proteins and lipids, like milk, the careful selection of solvents was essential for maintaining drop stability, considering the solubility of these components in non-polar solvents. The authors recommended pretreatment techniques, including adding salt and centrifugation, to enhance sample clarity and improve drop stability before the extraction process.

6. Artificial intelligence in SDME

The integration of artificial intelligence (AI) into analytical chemistry has the potential to revolutionize techniques such as SDME. AI can enhance the efficiency and accuracy of this extraction method by optimizing parameters such as solvent selection, extraction time, and temperature. Machine learning algorithms can analyze vast datasets to identify patterns and correlations that may not be immediately apparent, allowing for the development of predictive models that enhance the performance of SDME. For instance, AI could facilitate the selection of the most suitable ILs or solvents for specific analytes, thereby improving the extraction process and minimizing matrix

interference [128].

Despite the promising applications of AI in enhancing SDME, there remains a significant research gap in the literature. Currently, no experimental papers specifically explore the application of AI techniques to the SDME method. This absence highlights an opportunity for researchers to investigate how AI can be employed to optimize and innovate within this extraction technique. Addressing this gap could lead to the development of more robust, automated, and efficient methodologies for isolating drug analytes from complex sample matrices, particularly in environmental and clinical settings. By leveraging AI, researchers can not only improve the operational parameters of SDME but also expand its applicability across a wider range of analytes and sample types [129,130].

In conclusion, the intersection of AI and SDME presents an exciting frontier for research and development. As the demand for sensitive and selective analytical methods grows, harnessing the capabilities of AI could significantly enhance the efficacy of SDME, paving the way for future studies that fill the existing gaps in the literature. Emphasizing this need for exploration will encourage the scientific community to focus on innovative approaches that could transform the landscape of DA in complex matrices [130,131].

7. Conclusion and future perspective

In summary, the review highlights the evolving landscape of SDME techniques in pharmaceutical analysis, emphasizing the need for continued innovation and exploration to address existing challenges and unlock new opportunities in DA from complex sample matrices. In conclusion, using SDME techniques to isolate drug analytes from complex sample matrices presents a promising avenue for pharmaceutical analysis. Through advancements in technology and methodology, researchers have made significant strides in overcoming challenges related to matrix interference, achieving low detection limits, and enhancing the stability of acceptor drops. However, there remain important areas for future research and development.

Addressing the stability of acceptor drops, optimizing the combination of SDME with chromatography or spectroscopy, and improving detection limits for ultra-trace analysis of pharmaceuticals are paramount. Novel strategies such as the derivatization of pharmaceutical materials to reduce boiling points and increase absorptivity, as well as in-drop derivatization techniques, show promise in overcoming these challenges. Furthermore, optimizing the properties of acceptor solvents, particularly by exploring mixed solvent systems, emerges as a highly recommended solution. While previous studies predominantly focused on single solvent approaches, the potential benefits of mixed solvent acceptors warrant further investigation.

Currently, researchers face important challenges in ensuring the stability of acceptor drops, combining methods such as SDME and chromatography or spectroscopy, and achieving low detection limits for ultra-trace analysis of pharmaceuticals. Despite significant progress, novel strategies and methods are needed in these areas. To overcome these challenges, future research should focus on derivatizing pharmaceutical materials to reduce their boiling points and increase absorptivity or employ in-drop derivatization. However, the most recommended solution is optimizing the properties of acceptor solvents. Our review indicates that while most previous studies focused on single solvents, using mixed solvent acceptors may offer a better approach. The use of unconventional solvents such as ILs, surfactants, and eutectic solvents has significantly enhanced the efficiency of SDME. ILs, being salts in liquid form at room temperature, offer exceptional properties like negligible vapor pressure, high thermal stability, and tunable polarity, making them highly effective in extracting a wide range of analytes from complex matrices. Their ability to dissolve both polar and nonpolar compounds provides a distinct advantage over traditional solvents, particularly in pharmaceutical analysis. However, the high cost and potential toxicity of ILs are notable challenges. Similarly,

surfactants improve extraction efficiency by forming micellar systems that enhance analyte partitioning. Their amphiphilic nature allows for the effective isolation of both hydrophilic and hydrophobic drugs, though issues such as drop instability may impact reproducibility. Eutectic solvents, composed of hydrogen bond donors and acceptors, are environmentally friendly alternatives that offer adjustable viscosity and polarity, facilitating the extraction of various pharmaceutical compounds. Nonetheless, their higher viscosities and slower diffusion rates may affect drop stability and kinetics [132–134].

The application of a vacuum to reduce the pressure in the sample vial is another innovative approach in SDME. By creating a low-pressure environment, equilibrium is reached much faster than under atmospheric conditions, significantly reducing extraction times. This method is particularly effective for volatile compounds, as the reduced pressure enhances their evaporation rates, leading to improved analyte transfer into the microdrop. However, maintaining a stable vacuum and ensuring the structural integrity of the sample vial under reduced pressure can pose technical challenges. Additionally, this technique may be less effective for non-volatile analytes [135].

The in-vial temperature gradient technique in headspace SDME involves cooling the extractor drop while simultaneously heating the SS. This temperature differential increases the volatility of analytes in the sample vial while minimizing their loss from the cooled extractor drop. This approach is especially beneficial for semi-volatile drugs, as it maximizes extraction efficiency without compromising analyte integrity. However, precise temperature control is crucial, as improper adjustments can destabilize the microdrop or degrade heat-sensitive compounds. Despite these challenges, this method has demonstrated great potential for enhancing SDME performance in pharmaceutical analysis [136,137].

CRediT authorship contribution statement

Kimia Javar: Data curation, Formal analysis, Validation, Writing – original draft. **Amin Foroozandeh:** Formal analysis, Methodology, Validation, Visualization, Writing – original draft. **Mahsa Souri:** Data curation, Formal analysis, Software, Writing – original draft. **Hossein Salar Amoli:** Conceptualization, Formal analysis, Investigation, Supervision, Writing – original draft. **Mohammad Hasanzadeh:** Formal analysis, Investigation, Methodology, Supervision, Visualization, Writing – review & editing.

Declaration of competing interest

The authors confirm that they have no recognized financial conflicts or personal connections that might affect the results discussed in this paper.

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Data availability

The data that has been used is confidential.

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Abbreviation

Solid-phase microextraction: SPME
Solid-phase extraction: SPE
Organic solvent: OS
Ionic liquids: ILs
Organic Phase: OP
Sample solution: SS
Sample preparation: SP
Drug analysis: DA
Liquid-phase microextraction: LPME
Cold-induced aggregation microextraction: CIAME
Electro membrane isolation: EMI
Ionic liquids: IL
In situ solvent formation microextraction: ISFME
Liquid-gas-liquid microextraction: LGLME
Stir-bar microextraction: SBME
Direct immersion: DI
Headspace: HS
Gas chromatographic: GC
Homocysteine thiolactone: HTL
Atmospheric pressure: AP
Matrix-assisted laser desorption/ionization: MALDI
Mass spectrometry: MS
Atmospheric pressure matrix-assisted laser desorption/ionization mass spectrometry: (AP-MALDI-MS)
Capillary electrophoresis: CE
Muscimol: MUS
Psilocin: PSC
Gas chromatography flame ionization detection: GC-FID
Aminophenol isomers: APs
High-performance liquid chromatography: HPLC
Nonsteroidal anti-inflammatory drugs: NSAIDs
Single drop liquid-liquid-liquid microextraction: SD-LLLME
Methamphetamine: MAP
3,4-methylenedioxy- amphetamine: MDA
Tricyclic antidepressant drugs: TCAs
A-cyano-4-hydroxycinnamic acid: a-CHCA
Fluoroquinolones (FQs): FQs
Solvent microextraction (SME): SME
Dimethyl sulfoxide (DMSO): DMSO
Methyl methanesulfonate (MMS): MMS
Ethyl methanesulfonate: EMS
Isopropyl methanesulfonate: IPMS
Active Pharmaceutical ingredients: APIs
Ionic liquid-based immersed droplet microextraction: IL-IDME
Natural deep eutectic solvent: NADES
Low-density solvent-based solvent-demulsification dispersive liquid-liquid microextraction: LDS-SD-DLLME